Sentiment analysis of product reviews

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Contents

Results

Dataset

Preprocessing

Model

Pipeline

References

Results

The test set holds 6,000 reviews that were not used in training.

The lexicon-based scorer (VADER) gave the same label as the human annotators for 4,380 of the 6,000 reviews, an accuracy of 73.0 percent.

The logistic regression model on TF-IDF features gave the same label as the human annotators for 5,010 of the 6,000 reviews, an accuracy of 83.5 percent.

The macro F1 score was 0.71 for the lexicon-based scorer and 0.82 for the logistic regression model.

Neutral reviews had the lowest recall in both methods: 0.48 for the lexicon-based scorer and 0.61 for logistic regression.

Dataset

The data is a public collection of 30,000 English product reviews from an online marketplace, each with a star rating from 1 to 5.

Reviews with 4 or 5 stars were labelled positive, 3 stars neutral, and 1 or 2 stars negative.

The split was 24,000 reviews for training and 6,000 for testing, stratified by label.

Class counts in the training split: 11,200 positive, 6,100 neutral and 6,700 negative.

Preprocessing

Text was lower-cased and HTML tags and web addresses were removed.

Repeated punctuation was collapsed to a single mark.

Stop words were removed with the NLTK English list, except for no, not and nor, which carry sentiment.

Words were turned into lemmas with the WordNet lemmatizer.

Features were TF-IDF weights over word unigrams and bigrams, limited to the 50,000 most frequent terms.

Model

Two methods were run on the same test set.

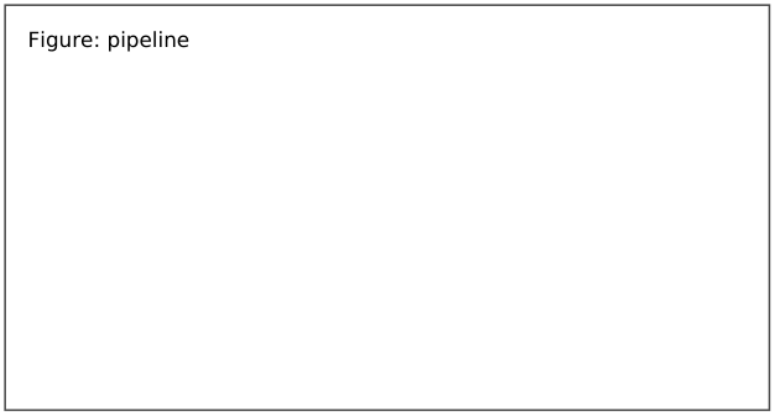
Method one is VADER, a rule-based scorer that sums word valence scores from a lexicon and applies rules for negation and intensifiers. Scores above 0.05 are positive and below -0.05 are negative.

Method two is a multinomial logistic regression with L2 regularisation, regularisation strength C = 4, trained with the saga solver for up to 300 iterations.

Class weights were set to balanced to account for the smaller neutral class.

Pipeline

Figure: pipeline



References

[1] Jurafsky, D. and Martin, J. H. Speech and Language Processing (3rd edition draft). Pearson.

[2] wikipedia sentiment page

[3] towardsdatascience article on VADER

[4] scikit-learn documentation, logistic regression page